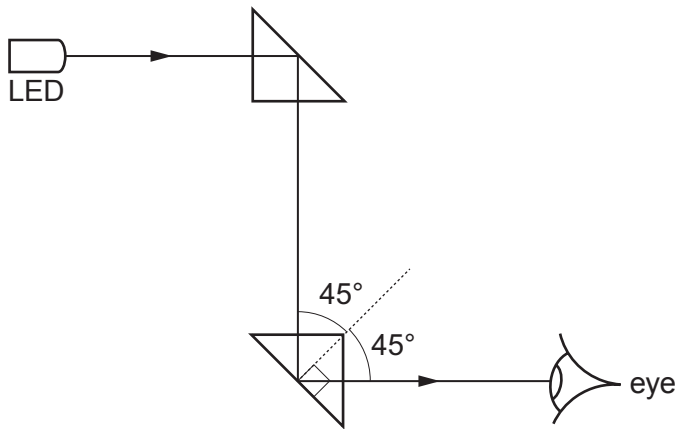


6. (a) A student sets up a simple periscope consisting of two glass prisms ($n_{\text{glass}} = 1.52$). She tests it by using it to view a light emitting diode (LED) as shown.



- (i) Give the full name of the process by which the light changes direction in the prisms. [1]
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- (ii) The student wonders whether the periscope will still work properly if the lower prism is surrounded by water ($n_{\text{water}} = 1.33$). Determine whether or not it will, giving your reasoning clearly. [3]
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- (b) (i) A particular monomode fibre has a core of refractive index 1.515. Calculate the time it takes for a pulse of light to travel through 1.50 km of the fibre. [3]
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(ii) Explain the advantage of a monomode fibre over a fibre with a much thicker core, for the transmission of a rapid stream of data. [3]

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Examiner
only

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